

HopTF 48-Variant Panel Validation

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Question: after removing ambiguous variants such as GATA2 R362Q from the primary validation set, do high-confidence deleterious TF missense variants reduce HopTF-predicted transport response more than high-confidence benign variants?

Panel: 24 benign / likely benign variants and 24 pathogenic / likely pathogenic variants. Each class spans 24 and 24 genes, respectively. All locked variants were coordinate-checked against local responder isoform sequences with n_cells ≥ 5 .

Evaluator: strict frozen HopTF endpoint evaluator status ok; 48 / 48 variants predicted.

Main result: deleterious variants shift more negative on average than benign variants, but the separation is modest under the current PCA-centroid OT-CFM checkpoint. Median percent delta is -0.381% for benign variants and -1.818% for deleterious variants. Mean percent delta is -0.135% for benign and -2.653% for deleterious. AUROC for deleterious classification using -percent_delta is 0.578.

Interpretation: the result supports weak directional sequence sensitivity but is not decisive validation. The benign panel behaves as expected for response stability: 95.8% of benign variants remain within +/-5% response change. Deleterious variants are only modestly enriched for response reductions: 54.2% have negative response delta.

Curation And Mapping Traceability

Source: ClinVar variant_summary release filtered on GRCh38 single-nucleotide missense substitutions in local HopTF responder TF genes.

Filtering: excluded VUS, conflicting interpretations, no-assertion/no-classification labels, risk/association/pharmacogenomic/protective labels, truncations, non-missense substitutions, coordinate mismatches, and variants outside local isoform sequence ranges.

ClinVar scan counts: raw rows 8,978,989; GRCh38 rows 4,455,872; SNV rows 4,133,147; local-gene rows 118,402; allowed clinical-significance rows 36,797; clean candidates 3,300.

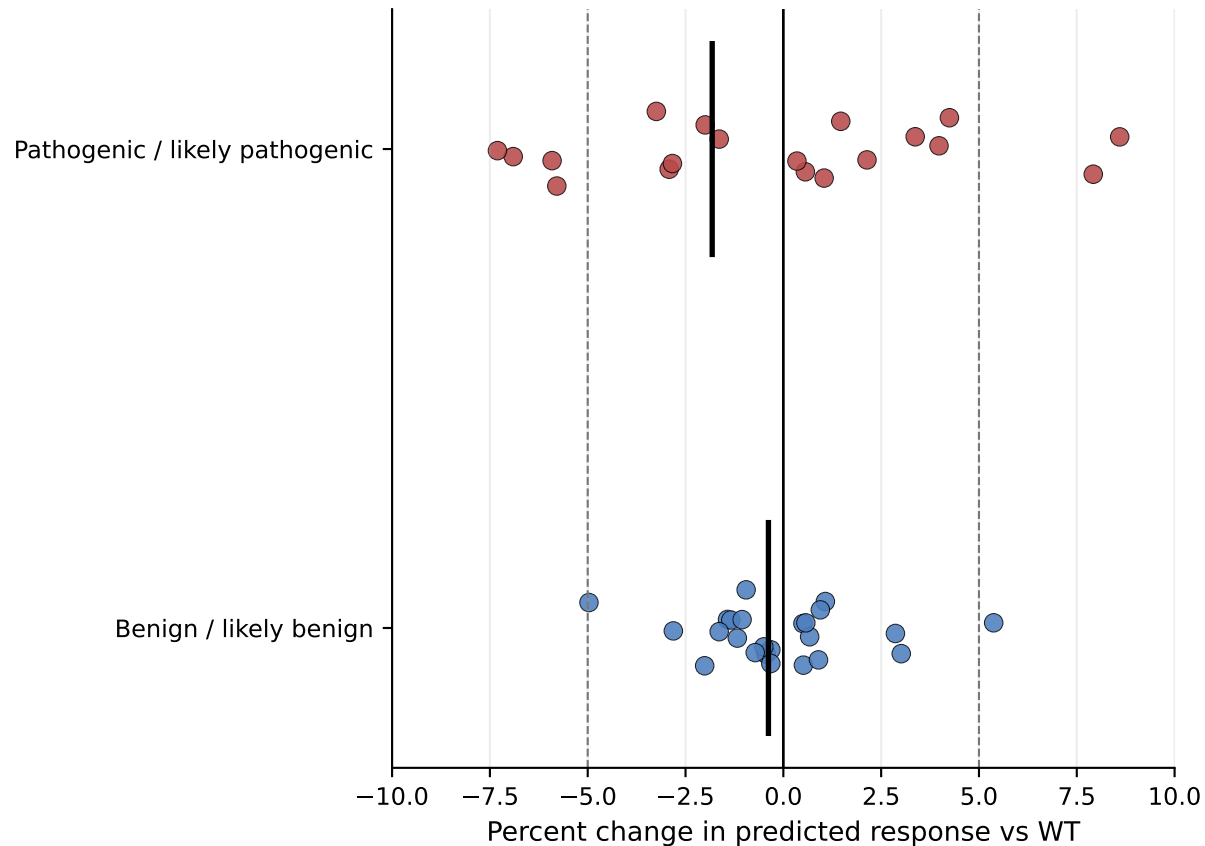
Coordinate mapping: 4,505 mapped rows after checking local responder isoforms. Mapping status counts were {'ok': 2906, 'coordinate_mismatch': 1225, 'out_of_range': 374}.

Locked panel: 48 variants total, with 24 benign and 24 deleterious variants. Each class covers 24 and 24 genes, respectively.

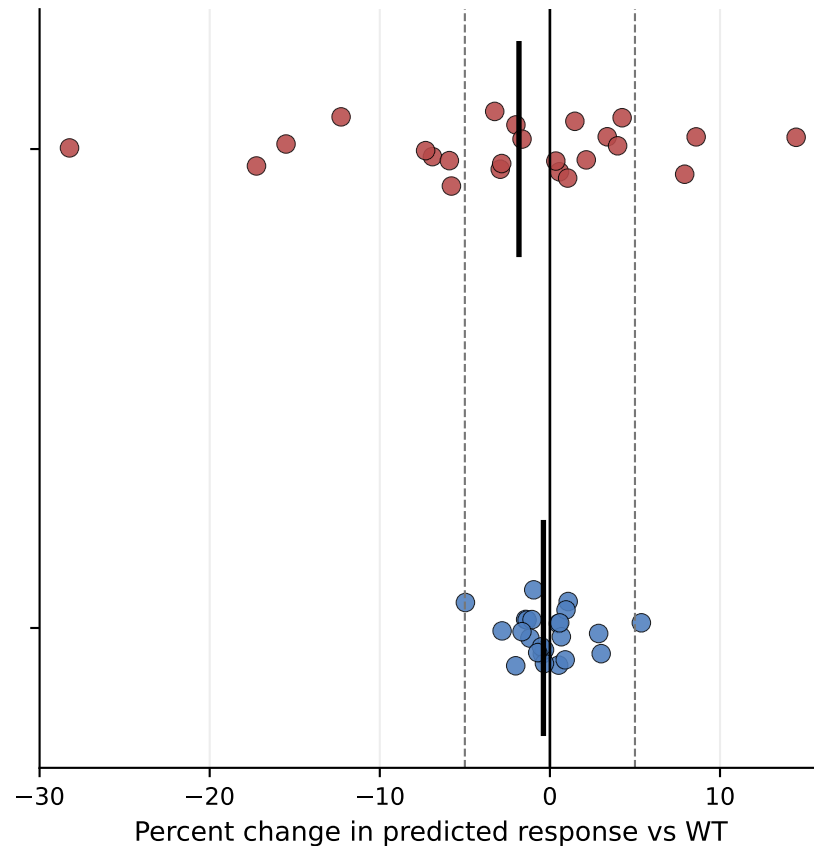
Caveat: ClinVar pathogenicity is a disease-variant label, not always a direct assay of TF perturbation response in this cellular system. This is therefore a cleaner validation panel than ambiguous hand-picked variants, but not a perfect mechanistic benchmark.

Deleterious variants shift more negative on average, but overlap is substantial

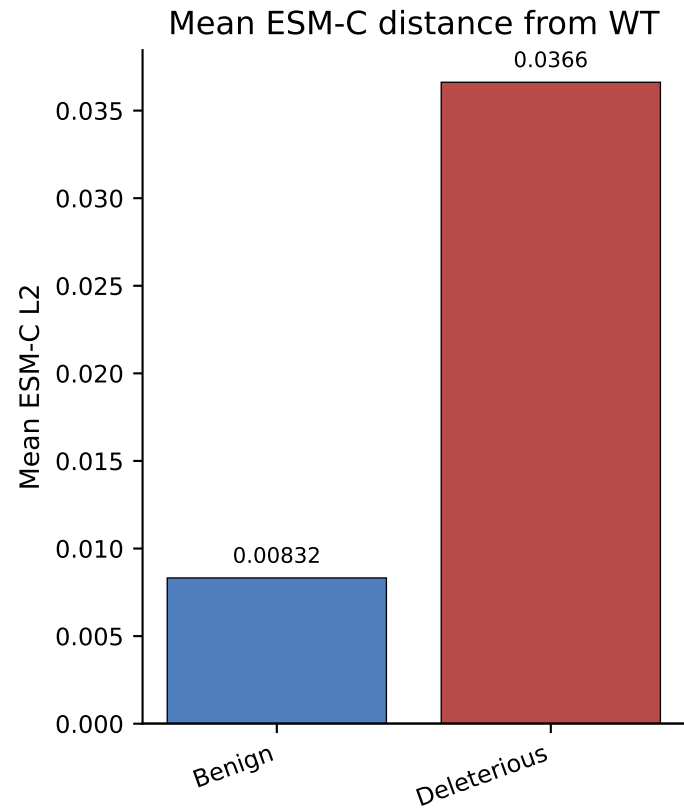
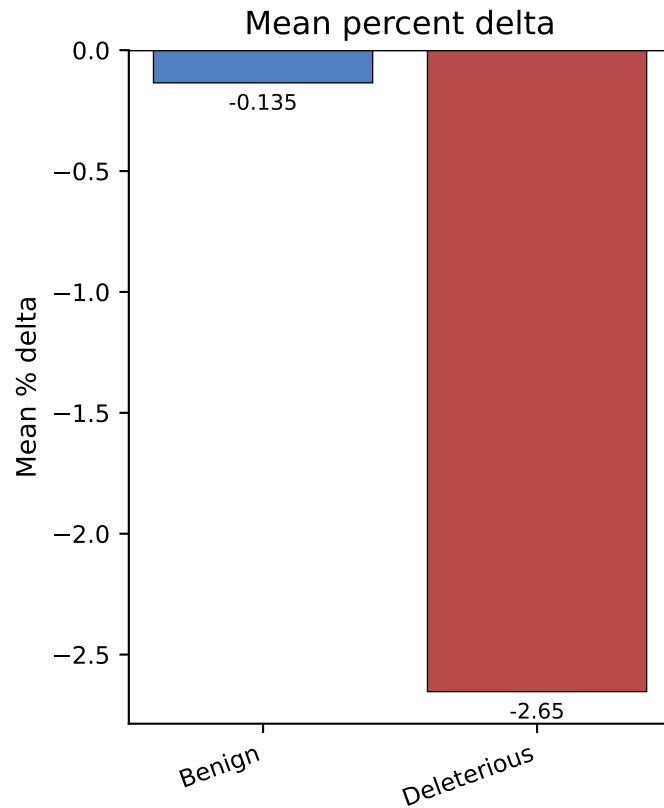
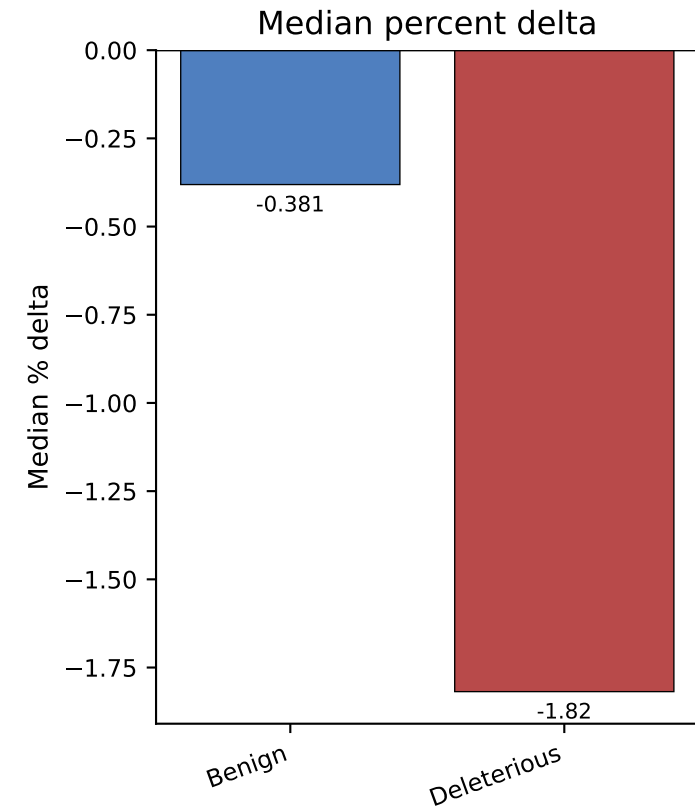
Zoomed view: main response range



Full range including tails



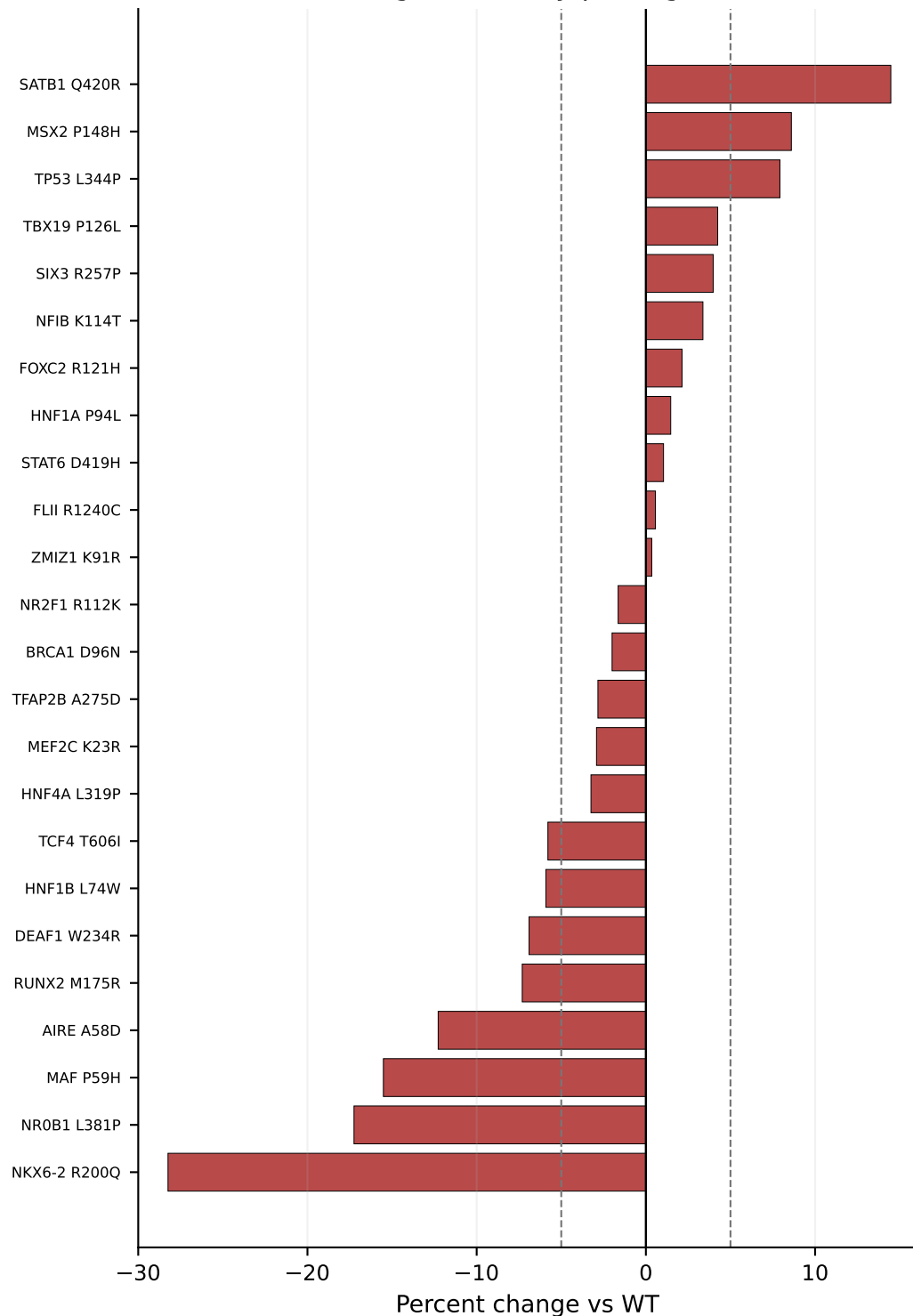
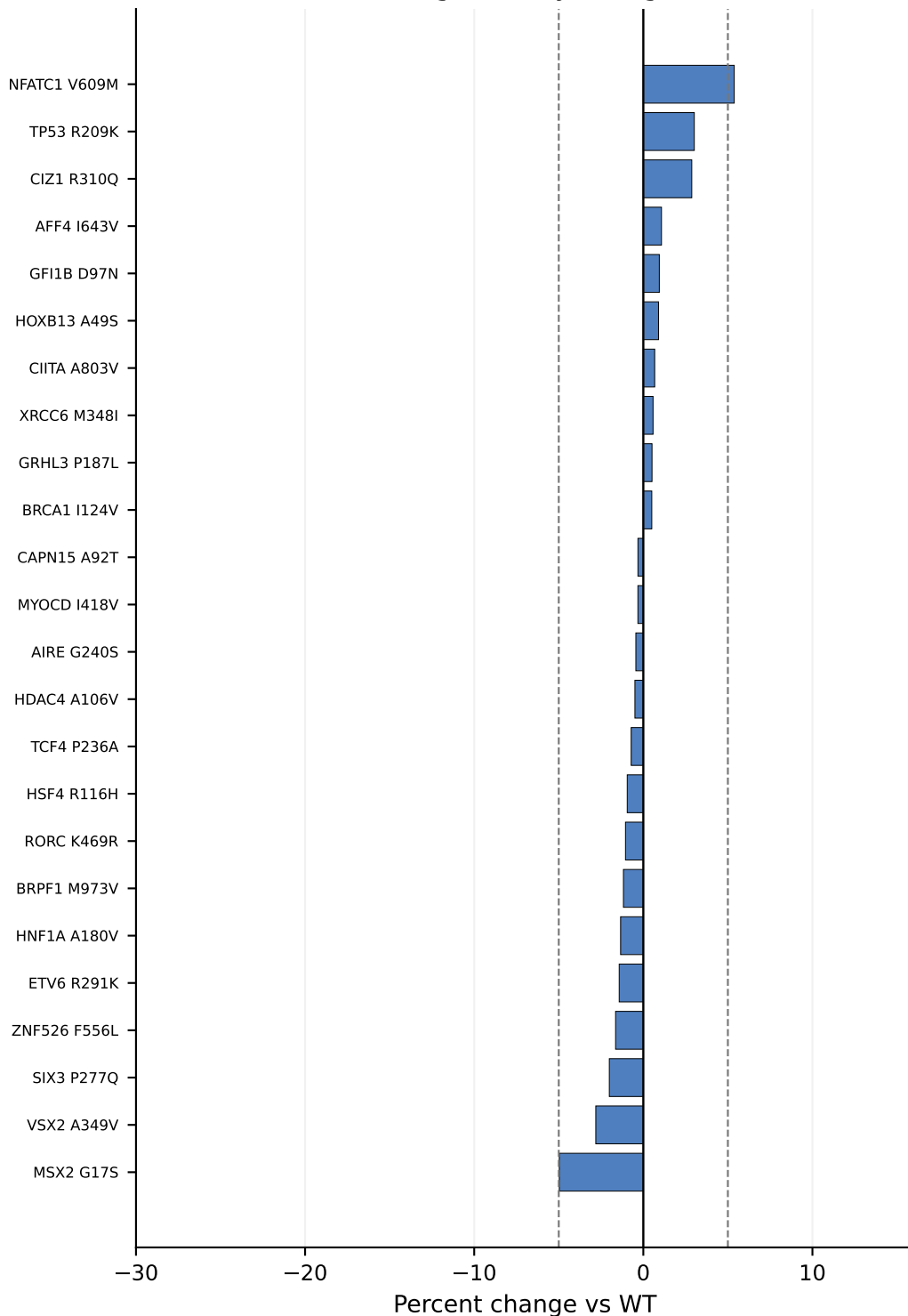
Aggregate validation metrics



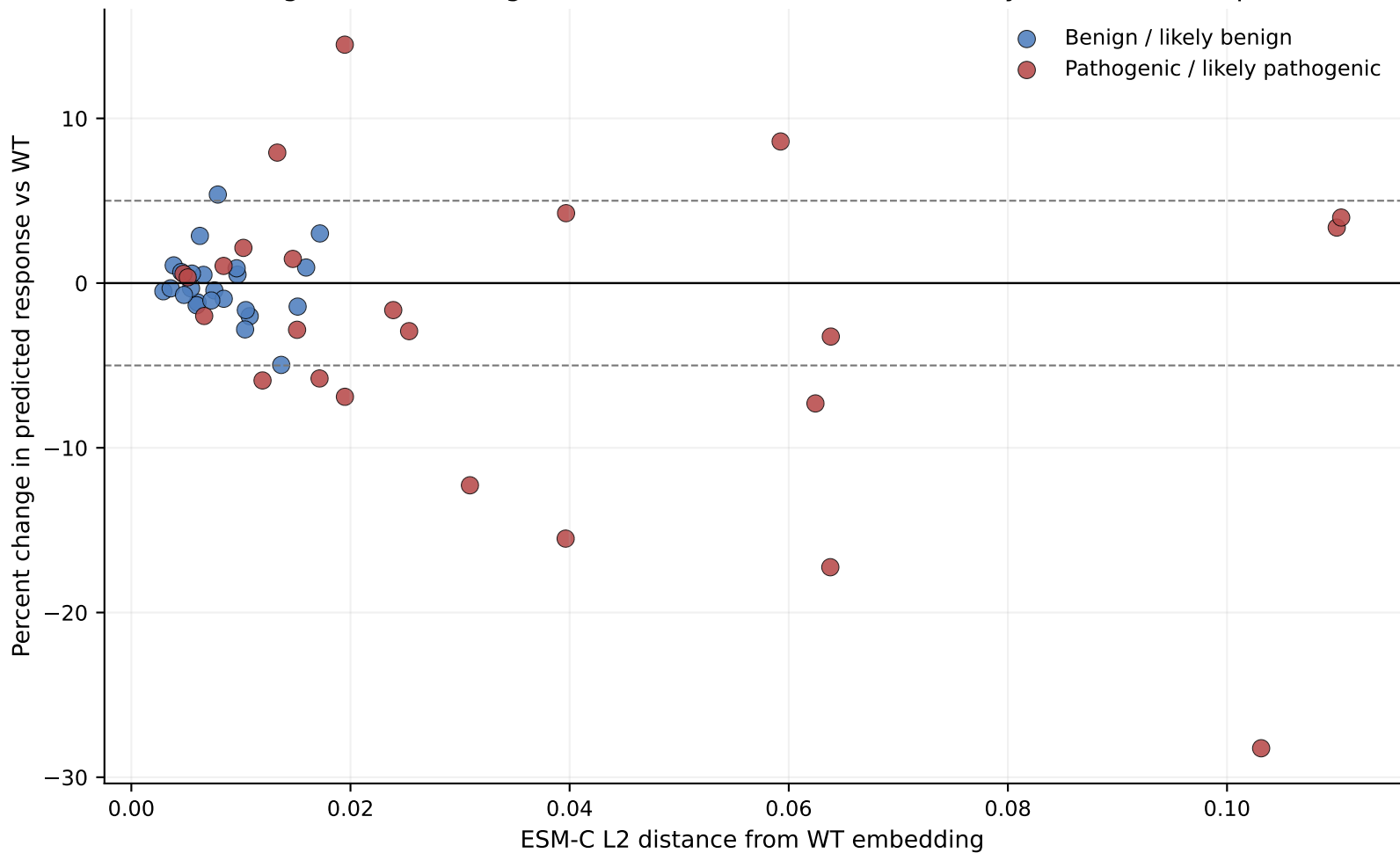
Per-variant response changes, sorted within class

Benign / likely benign

Pathogenic / likely pathogenic



Embedding distance is larger for deleterious variants and may confound interpretation



Largest response decreases and increases

Class	Gene	Mutation	ClinVar label	Review	% delta	Delta L2	ESM dist.
deleterious	NKX6-2	R200Q	Pathogenic	criteria provided, single submitter	-28.2	-140	0.103
deleterious	NR0B1	L381P	Likely pathogenic	criteria provided, single submitter	-17.2	-93.2	0.0638
deleterious	MAF	P59H	Likely pathogenic	criteria provided, single submitter	-15.5	-88.2	0.0396
deleterious	AIRE	A58D	Likely pathogenic	criteria provided, single submitter	-12.3	-75.4	0.0309
deleterious	RUNX2	M175R	Likely pathogenic	criteria provided, single submitter	-7.31	-24.7	0.0624
deleterious	DEAF1	W234R	Likely pathogenic	criteria provided, single submitter	-6.9	-36.4	0.0195
deleterious	HNF1B	L74W	Pathogenic	criteria provided, single submitter	-5.91	-56.3	0.012
deleterious	TCF4	T606I	Likely pathogenic	reviewed by expert panel	-5.79	-28.7	0.0172
deleterious	SATB1	Q420R	Likely pathogenic	criteria provided, single submitter	14.5	26.3	0.0195
deleterious	MSX2	P148H	Pathogenic	criteria provided, single submitter	8.6	36.8	0.0593
deleterious	TP53	L344P	Likely pathogenic	reviewed by expert panel	7.92	24	0.0133
benign	NFATC1	V609M	Likely benign	criteria provided, single submitter	5.38	23	0.00789
deleterious	TBX19	P126L	Pathogenic	criteria provided, single submitter	4.24	36.3	0.0397
deleterious	SIX3	R257P	Likely pathogenic	criteria provided, single submitter	3.98	41.5	0.11
deleterious	NFIB	K114T	Pathogenic	criteria provided, single submitter	3.37	26.7	0.11
benign	TP53	R209K	Likely benign	reviewed by expert panel	3.01	9.12	0.0172

This table shows tails from both directions. Several deleterious variants move in the wrong direction, which is why the overall AUROC is modest.

Bottom Line

This larger validation panel is cleaner than the initial hand-picked mutation set because it excludes ambiguous variants and balances high-confidence benign and deleterious ClinVar labels.

The result is not a strong validation win. The benign variants mostly behave as expected: 23 / 24 are within +/-5% predicted response change. Deleterious variants are shifted more negative on average, but only 13 / 24 reduce predicted response and the class separation is modest (AUROC 0.578).

The most defensible report statement is: HopTF endpoint predictions show weak directional sensitivity to high-confidence deleterious substitutions relative to benign substitutions, but the current PCA-centroid OT-CFM checkpoint does not robustly separate deleterious from benign variants. This strengthens the case for keeping claims bounded and for prioritizing beta/motif diagnostics and endpoint improvements before claiming biological validation.